

# AquaTrace Maritime Forensic Report

CASE REF: AT-REP-OS-042-2026

EXPLAINABLE MARITIME OIL SPILL DETECTION, TRACEBACK & VESSEL ATTRIBUTION | 2026 PS 143 • NTRO • CODE BLOODED

## 1. COVER PAGE & REFERENCE BLOCK

### INCIDENT IDENTIFIER:

OS-042

### STATUS / VERDICT:

HIGH CORRELATION (89.9/100)

### DETECTION TIME (UTC):

2026-09-27 14:32:00 UTC

### GEOGRAPHIC REGION:

Arabian Sea (Offshore Mumbai / ENR Route)

### ASSIGNED INVESTIGATOR:

Capt. A. Sharma (ICG HQ)

### DEMONSTRATION STATUS:

SYNTHETIC / DEMO MODE

FORENSIC NOTICE: This investigative brief establishes priority targets for Port State Control (PSC) verification and does not assert unilateral civil/criminal liability. Synthetic AIS and simulated metocean forcing parameterize the demonstration.

## 2. EXECUTIVE SUMMARY

On 2026-09-27 14:32:00 UTC, satellite radar observation (Sentinel-1C (SAR-C)) detected an anomalous surface slick of estimated area 14.85 km<sup>2</sup> (estimated volume ~420 m<sup>3</sup>) in the Arabian Sea (Offshore Mumbai / ENR Route). Reverse Lagrangian hydrodynamic hindcasting driven by simulated metocean forcing (currents: 0.85 kts @ 64°, winds: 16.5 kts @ 250°) reconstructed a release window between 08:45–10:15 UTC at centroid 18.26°N, 70.92°E.

Spatio-temporal correlation identified candidate MT AL-HIKMA (Panama, MMSI: 477291000) as having the highest correlation (Composite Evidence Score: 89.9/100, HIGH CORRELATION). Counterfactual forward release simulation matched the observed SAR slick geometry with 91.4% shape similarity and 84.2% spatial IoU (demonstration metrics).

## 3. INCIDENT OVERVIEW & GEOSPATIAL BOUNDARY

### SLICK CENTROID

18.42°N, 71.18°E

### ESTIMATED SURFACE AREA

14.85 km<sup>2</sup>

### ESTIMATED VOLUME (BONN)

~420 m<sup>3</sup> (~2642 bbl)

Geospatial boundary encompasses maritime shipping corridors inside India's Exclusive Economic Zone (EEZ). Search bounds are delimited by the Lagrangian envelope: Lat [18.17°N – 18.67°N], Lng [70.93°E – 71.43°E].

## 4. SATELLITE OBSERVATION & SAR SENSOR METRICS

The oil slick anomaly was captured via Sentinel-1 Synthetic Aperture Radar (SAR), which operates independently of solar illumination and cloud cover by detecting normalized radar cross-section (NRCS) damping caused by short-gravity wave suppression. Browser demonstration uses pre-vectorized SAR-derived geometry; raw Level-1 GRD preprocessing and neural segmentation represent production integration targets.

• Satellite Platform: Sentinel-1C (SAR-C)

• Polarization: VV + VH

• Incidence Angle: 38.4°

• Sensor Instrument: C-SAR IW (Interferometric Wide Swath)

• Spatial Resolution: 10 meters

• Scene ID: S1C\_IW\_GRDH\_1SDV\_20260927T143210\_018291\_023FA1\_4A8F

## 5. SPILL CHARACTERISATION & BONN THICKNESS ESTIMATE

### MAJOR AXIS (LENGTH)

11.2 km

### MINOR AXIS (WIDTH)

1.8 km

### ASPECT RATIO & ORIENTATION

6.22 : 1 @ 62°

Bonn Agreement Appearance Formulation: Volume is estimated empirically based on appearance metallic sheen convention): V "H Area (14.85 km²) × Thickness (28.3 ¼m) "H 420 m³. Weatherin (water cut ~18%). Note: Demonstrative empirical estimate; not equivalent to direct laboratory volumetric measurement.

6. ORIGIN-TIME RECONSTRUCTION & AGE WINDOW

Origin is expressed as nested spatial uncertainty envelopes representing modelled spatial uncertainty around the reverse-advection origin estimate (not statistically calibrated Bayesian credible intervals):

| P50 CORE ORIGIN (50%) | P80 SEARCH ENVELOPE (80%)    | P95 DISPERSION BOUND (95%) |
|-----------------------|------------------------------|----------------------------|
| ±4.2 km (Apex core)   | ±8.6 km (Kinematic corridor) | ±14.8 km (Maximum lateral) |

Modelled Release Window: 08:45 – 10:15 UTC (estimated age window: 4.5–6.0 h prior to SAR overpass). Origin centroid: 18.26°N, 70.92°E.

7. ANALYTICAL LAGRANGIAN HINDCAST

Backward trajectory reconstruction uses client-side analytical vector advection parameterizing surface current forcing and atmospheric wind leeway:

- Governing Transport Equation:  $U_{total} = U_{current} + 0.033 \times U_{wind}$
- Surface Current Field ( $U_{curr}$ ): 0.85 kts @ 64° (parameterizing CMEMS advection)
- Atmospheric Wind Field ( $U_{wind}$ ): 16.5 kts @ 250° (parameterizing ERA5 wind stress)
- Wind Leeway Factor ( $\pm$ ): 3.3 % standard empirical coefficient
- Traceback Horizon: ~5.2 hours reverse advection to origin apex

8. HYDRODYNAMIC DISPERSION FORECAST

Forward hydrodynamic trajectory modeling projects slick movement across 6h, 12h, and 24h operational response horizons:

- T+6h Projected Offset: ~5.8 nm @ 072°
- T+12h Projected Offset: ~12.4 nm @ 075°
- T+24h Projected Offset: ~23.8 nm @ 080°
- Shoreline Intercept Assessment: Low risk; slick advecting parallel to continental shelf break.
- Production Target: Full OpenDrift / OpenOil integration with weathering, evaporation, and vertical droplet entrainment.

9. AIS TELEMETRY RECONSTRUCTION

Vessel traffic reconstruction ingests synthetic scenario AIS telemetry following the MarineCadastre-compatible schema over the temporal window [Overpass - 6h to Overpass]. Trajectories are filtered for spatial intersection with the P80/P95 origin envelopes.

Dark Vessel Screening Alert: SAR CFAR detector identified an uncooperative metallic contact (SAR-DV-01) with length ~38m (RCS: 24.5 dB). Nearest AIS broadcast was 4.8 nm distant. Classified as potential uncooperative contact; candidate correlation continues for reporting fleet.

10. MULTI-CHANNEL CANDIDATE COMPARISON TABLE

The six-channel evidence weighting synthesizes multi-source forensic data into a single Composite Evidence Score (Weights sum to exactly 1.00: Satellite 20%, Drift 25%, AIS 20%, Behaviour 10%, Counterfactual 20%, Historical Context 5%):

| CANDIDATE VESSEL           | SAT(20%) | DRIFT(25%) | AIS(20%) | BEH(10%) | SIM(20%) | SCORE | STATUS |
|----------------------------|----------|------------|----------|----------|----------|-------|--------|
| #1 MT AL-HIKMA (Panama)    | 92%      | 94.2%      | 88.5%    | 81.3%    | 91.4%    | 89.9  | HIGH   |
| #2 PACIFIC GLORY (Liberia) | 78%      | 66.5%      | 72%      | 45%      | 58.6%    | 66.9  | LOW    |
| #3 NORDIC RUNNER (Norway)  | 35%      | 22%        | 38%      | 20%      | 24.1%    | 31.5  | LOW    |

Operational weighting notice: Weights represent operational decision-support criteria; not statistically calibrated against empirical ground-truth base rates. Highest-Ranked Candidate does not establish judicial guilt.

11. IN-SILICO COUNTERFACTUAL RELEASE SIMULATION

Hypothesis Evaluation: "If candidate MT AL-HIKMA released oil along its reconstructed CPA track at release time T\_cpa, would forward hydrodynamic transport match the observed SAR slick geometry?"

|                     |                    |                      |                  |
|---------------------|--------------------|----------------------|------------------|
| SPATIAL IoU OVERLAP | HAUSDORFF DISTANCE | TRAJECTORY ALIGNMENT | SHAPE SIMILARITY |
| 84.2%               | 1.15 km            | 94.2%                | 91.4%            |

Demonstration Metric Disclosure: Counterfactual metrics (IoU: 84.2%, Hausdorff: 1.15 km, Shape Similarity: 91.4%) are demonstration scenario values illustrating verification pipeline architecture.

12. EVIDENTIARY PROVENANCE GRAPH

Directed Acyclic Graph (DAG) traces evidentiary transformation from raw sensor downlink to candidate ranking:

[ SCENE - S1C - 143210 ] % % % % [ SEGMENTATION - MODULE ] % % % % [ SLICK - POLY - OS - 042 ]  
%  
% % % % [ HINDCAST - ADVECTION ] % % % % [ ORIGIN - P50 - CENTROID ]  
%  
% % % % [ AIS - CORRIDOR - FILTER ] % % % % [ CANDIDATE - 477291000 ]  
%  
% % % % [ COUNTERFACTUAL - TEST ] % % % % [ COMPOSITE - SCORE : 89.9 ]

13. KINEMATIC FEASIBILITY & CONTINUITY ANALYSIS

Speed Profile Assessment: MT AL-HIKMA transited the sector with baseline cruise of 14.2 knots, decelerating to 8.4 knots for 42 minutes inside the reconstructed release corridor before resuming nominal speed.  
Heading Alignment: Course at CPA (064°) aligns within 2° of slick major axis orientation (062°).  
Investigative Notice: Kinematic speed changes indicate operational anomalies requiring investigator review; they do not by themselves establish vessel manipulation or wrongdoing.

14. UNCERTAINTY BOUNDS & SCIENTIFIC LIMITATIONS

- Look-Alike Screening: Biogenic films and internal waves mimic oil damping; screened against wind threshold (3–12 m/s).
- Drift Uncertainty: Sub-mesoscale eddies and Stokes drift introduce ~12% lateral uncertainty over 6h traceback horizons.
- AIS Line-of-Sight: Terrestrial VHF AIS is subject to range limits (>35 nm offshore).
- Evidence-Based Abstention: Where counterfactual IoU is <50% or candidate separation is ambiguous, AquaTrace declares INCONCLUSIVE.

15. INVESTIGATION FINDING & CORRELATION STATUS

FINAL FINDING: HIGH CORRELATION — HIGHEST-RANKED CANDIDATE IDENTIFIED  
Candidate: MT AL-HIKMA (MMSI: 477291000, Flag: Panama)  
Composite Evidence Score: 89.9 / 100  
Correlation Tier: HIGH CORRELATION  
Notice: Findings constitute targeted investigative intelligence for maritime authorities and do not represent judicial liability confirmation.

16. IMMUTABLE AUDIT TRAIL & RECORD INTEGRITY

Chronological event ledger recording analysis pipeline state transitions with demonstration provenance identifiers:

- [2026-09-27 14:32:10 UTC] Ingested Sentinel-1C SAR Scene | System: Automated Ingestion Worker | Ref: e3b0c44298fc1c14...
- [2026-09-27 14:32:38 UTC] SAR Oil Slick Segmentation (SegFormer-B4) | System: Inference Engine #03 | Ref: 4f53cda18c2baa0c...
- [2026-09-27 14:33:12 UTC] Ensemble Backward Drift Hindcasting | System: Simulation Node #01 | Ref: 7c6a54d092c10b23...
- [2026-09-27 14:33:45 UTC] Spatio-Temporal AIS Correlation | System: Attribution Service | Ref: 98d3c11492ba0c03...

17. INGESTION DATA PROVENANCE MATRIX

- Satellite SAR Source: ESA Copernicus Sentinel-1C C-SAR (Demonstration geometry)
- Meteorological Forcing: Simulated wind vectors parameterizing ERA5 atmospheric forcing
- Ocean Hydrodynamics: Simulated current fields parameterizing CMEMS hydrodynamic advection
- AIS Telemetry Feed: Synthetic transponder records formatted to MarineCadastre standards
- Benchmark Corpus: JRC Zenodo Sentinel-1 SAR Oil Spill Dataset (Reference training target)

18. ANALYTICAL ATTRIBUTION METHODOLOGY

- Lagrangian Advection Formula:  $U_{total} = U_{current} + 0.033 \times U_{wind}$  (calculated client-side via vector decomposition).
- Spatial IoU Formulation:  $IoU = \frac{Area(S_{observed} \cap S_{simulated})}{Area(S_{observed})}$
- Candidate Evidence Synthesis:  $Score = 0.20 \cdot Sat + 0.25 \cdot Drift + 0.20 \cdot AIS + 0.10 \cdot Behaviour + 0.20 \cdot Counterfactual + 0.05 \cdot History$ .
- Abstention Criterion: Enforced when look-alike risk > 80% or candidate score separation < 10%.

19. DEMONSTRATION MODE & REALITY DISCLOSURES

ZERO-FABRICATION SCIENTIFIC DISCLOSURE: AquaTrace operates as a functional deterministic demonstration architecture for Smart India Hackathon 2026 PS 143. All vessel tracks are synthetic scenario data structured to MarineCadastre standards. Metocean vectors simulate ERA5 and CMEMS values. Production ML inference and full OpenDrift hydrodynamic integration are documented architectural targets.

20. OFFICIAL INVESTIGATOR SIGN-OFF & VERIFICATION PROTOCOL

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>LEAD INVESTIGATOR:</b> Capt. A. Sharma (ICG HQ)</p> <p>Maritime Disaster Management Directorate // NTRO Forensic Division</p> <p>AquaTrace Platform v2.4 • Report Generated: 2026-09-27T09:29:10.932Z</p> | <p><b>EVIDENTIARY VERIFICATION STATUS:</b></p> <p><b>VALIDATED FOR PORT STATE CONTROL TARGETING</b></p> <p>SHA-256 Provenance ID: e3b0c44298fc1c149afb4c8996fb924</p> |
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